

John Wilson Cheek

jwcheek03@gmail.com

Raleigh, NC 27606 | 804-994-7360 | jwcheek03.github.io/portfolio | linkedin.com/in/jwcheek03

Education

B.S. Computer Science: Game Development Concentration | North Carolina State University, Raleigh, NC **May 2026**
Minor in Creative Writing | University Honors Program | Dean's List All Semesters **GPA: 3.9/4.0**
Summer Study Abroad | University of Oxford, Oxford, UK **Jun 2024 – Jul 2024**

Relevant Coursework: Game Engine Foundations, Computer Graphics, Introduction to Artificial Intelligence, Applied Differential Equations, Communication for Engineering & Technology, Software Engineering, C and Software Tools

Skills

Languages: Java, C, C++, C#, Python, HTML, CSS, Prolog, Unreal Engine Blueprints, R
Tools: Unreal Engine, Visual Studio, GIT, Perforce, Jenkins, Maven, Adobe Suite, Office Suite, Blender

Project Experience

2D Game Engine | Work in Progress: Expected Completion November 2025

- Developed a custom video game engine in C++ using the SDL3 library specialized for card games
- Implemented core systems including networking, physics, input handling, and event management
- Applied software engineering best practices, including object-oriented design and memory management

The Impossible Winter | Open World RPG created in Unreal Engine 5.6 using Blueprints and C++

- Implemented core mechanics with C++ and integrated Blueprint Scripting for UI elements
- Designed and prototyped a large open world using UE 5.6 landscape, foliage, and modeling tools
- Adapted one of my short stories into an interactive gameplay narrative with a focus on environmental storytelling

Artificial Intelligence Problem Solving | Java and Prolog solutions to simple AI challenges

- Programmed an agent in Java to play connect four using the Minimax Algorithm with Alpha-Beta pruning
- Implemented enhanced A* pathfinding in Java to enable an AI agent to navigate a puzzle environment
- Designed and implemented a query-based movie recommendation algorithm in Prolog

Work Experience

Event Operations Technician | North Carolina State University Theatre | Raleigh, NC Sep 2022 – Present

- Designed lighting scenes for student organizations, local companies, and touring shows in a 700-seat venue
- Facilitated communication between board operators and 5-20 performers/show crews to ensure successful events

Stage Manager | North Carolina State University Theatre | Raleigh, NC Sep 2024 – Jun 2025

- Served as the central communication point between the director, show designers, and over 20 company members
- Managed scheduling, cue timing, and rehearsal documentation during 370 cumulative hours of production
- Adapted to real-time changes and issues during live performances and instructed crew members how to respond

Research Assistant | North Carolina State University | Raleigh, NC Nov 2024 – Feb 2025

- Gathered and analyzed research survey results from 122 industry professionals at over 24 companies
- Categorized data points into 21 key competencies and 10 factors that describe effective product managers

Campus Involvement

Video Game Development Club | Club Member | Raleigh, NC Sep 2025 – Present

- Attended club regular club meetings and participated in club-sponsored workshops and game jams
- Assisted other club members in learning the basics of commercial game engines

Alpha Psi Omega– Psi Kappa Cast | Fundraising Committee Chair | Raleigh, NC Feb 2025 – Present

- Organized monthly activities to reach out to prospective members and raise funds for the organization
- Ran outreach and fundraising events over a total of 40 cumulative volunteer hours

Certifications

Epic Games: Game Design Professional Certificate

Expected completion: Oct 2025